

SARTHAK ARSUL

+91 9168342306 sarthakarsul18@gmail.com [LinkedIn](#) [GitHub](#) [Portfolio](#) Pune, Maharashtra - 411057

Profile Summary

AI Engineer with hands-on experience building **LLM-powered AI Agents, RAG systems, and autonomous workflows** using **Python, LangChain, LangGraph, MCP, FastAPI, and Hugging Face**. Experienced in designing **tool-calling agents, semantic search, vector databases, prompt engineering, and production-ready AI applications**. Strong foundation in Machine Learning, NLP, REST APIs, Docker, Git, and scalable backend development.

Technical Skills

Generative AI & LLM Engineering: LangChain, LangGraph, Model Context Protocol (MCP), RAG (Retrieval-Augmented Generation), Hugging Face, Groq, Vector Databases (FAISS), Prompt Engineering, Embeddings, Semantic Search, AI Agents, Tool/Function Calling

Programming & Backend Development: Python, FastAPI, Flask, Django (basic), REST API, SQL

Machine Learning & Data Science: Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Regression, Classification, Clustering, Neural Networks, Deep Learning, NLP

Databases, DevOps & Tools: MySQL, PostgreSQL, Docker, MLflow, GitHub, CI/CD, Postman, AWS (Basic), Power BI, Matplotlib, Seaborn.

Projects

AI Movie Research Agent - MCP-based AI Agent

[GitHub Link](#)

- Built an end-to-end **AI Agent** using **Python, LangGraph, LangChain, Groq, and Model Context Protocol (MCP)** for natural-language movie research and dynamic tool calling.
- Developed a custom **MCP Server using FastMCP** and **MCP Client with stdio transport**, enabling dynamic tool discovery and execution through **JSON-RPC-based tools/list and tools/call**.

Self-Healing Data Pipeline - AI Agent for Automated Data Quality & Schema Repair

[Live Demo](#)

[GitHub Link](#)

- Built an **autonomous data-healing agent** using LangGraph, LangChain, and Groq LLM (Llama-4 Scout) to detect and auto-correct CSV schema mismatches, missing values, duplicates, and invalid records.
- Designed a **RAG pipeline with FAISS vector store and Hugging Face embeddings** to ground LLM-based column mapping in business-specific data governance rules.
- Implemented a stateful, **conditional retry loop** (LangGraph graph with conditional edges) enabling iterative self-correction without manual intervention.
- Handled LLM output-parsing failures with graceful fallback to prevent pipeline crashes.
- Tested pipeline against CSVs with up to **5,000** rows across 4 data-quality issue types
- Deployed** via an interactive **Streamlit** dashboard with real-time before/after data quality metrics.

DocMind - AI-Powered RAG Chatbot for PDF Documents

[Live Demo](#)

[GitHub Link](#)

- Built an end-to-end **Retrieval-Augmented Generation (RAG)** chatbot using **LangChain, Groq (Llama 4 Scout), FAISS, and Hugging Face Embeddings** to enable context-aware question answering over PDF documents.
- Engineered a **LangChain (LCEL)** retrieval pipeline with semantic search, optimized document chunking, and prompt engineering to generate grounded responses while minimizing hallucinations.
- Implemented **content-hash-based FAISS caching** to reuse previously processed PDF embeddings, significantly reducing document reprocessing time on repeated uploads.
- Evaluated the RAG pipeline using **20+ document-grounded test queries**, achieving **9.4/10 retrieval quality** while minimizing unsupported responses through prompt grounding and source-based retrieval.
- Developed a custom **Streamlit** interface with real-time response streaming, source citations, and a modular architecture for improved scalability and maintainability.

- AI Blog Writer – Multi-Agent Content Generation System (CrewAI + Groq)

[GitHub Link](#)
- Built a multi-agent content pipeline using **CrewAI** with three specialized agents (Researcher, Writer, Editor) orchestrated sequentially to autonomously research, draft, and edit blog articles end-to-end.
 - Integrated **Groq-hosted LLM** (llama-4-scout-17b-16e-instruct) for fast inference across all agent roles, with each agent's output feeding into the next stage of the pipeline.
 - Automated the end-to-end research-to-publication workflow, reducing manual content creation effort.

Experience

Python AI Intern

Docboyz | Nov 2025 – May 2026

- Built an **end-to-end Video KYC (VKYC) platform** using **FastAPI and Django**, covering user onboarding, live video capture, session management, recording, backend verification, and audit-ready workflows.
- **Deployed** and maintained the Video KYC (VKYC) platform on an **AWS EC2 Linux server using PuTTY (SSH) and WinSCP**, managing application deployment, server configuration, and production updates.
- Developed a **PAN Card data extraction system** using **PaddleOCR**, enabling automated identity data extraction and validation.
- Integrated third-party **Insurance APIs** (including **HDFC** and **Hizuno**) to automate policy verification, customer data exchange, and insurance workflow processing.
- Developed **Playwright-based web automation and web scraping** solutions to extract, validate, and process structured data from web portals, reducing manual effort and improving operational efficiency.

Certifications

Data Science – *Ethan’s Tech*

Python Programming – *Ethan’s Tech*

Education

Deogiri College, Sambhajinagar - BCA – 73.43%	Jun 2020 – May 2023
Balbhim College, Beed - 12 th (HSC) – 63.38%	Jun 2019 – May 2020